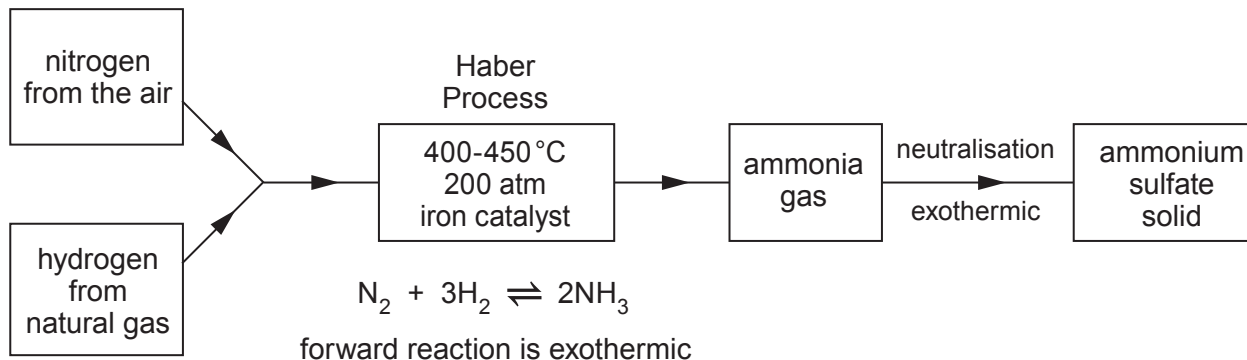


7. The flow diagram outlines the manufacture of ammonia by the Haber Process followed by the production of ammonium sulfate by neutralisation.



- (a) Explain the choice of temperature used in the Haber Process and the reason why a catalyst is used. [3]

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- (b) Powdered ammonium sulfate is industrially formed by spraying sulfuric acid into a reaction chamber filled with ammonia gas. The exothermic reaction that occurs evaporates all water present in the system, forming a powdery salt.

Write a balanced symbol equation for this reaction. [2]

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only

(c) In the laboratory ammonium sulfate is prepared by reacting ammonium hydroxide solution with dilute sulfuric acid.

Describe the titration method for making pure crystals of ammonium sulfate in the laboratory. Explain each stage of your method. [6 QER]

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